

Ali Meski

519-584-5194 | ameski@uoguelph.ca | www.linkedin.com/in/ali-meski | Ali's Portfolio

EDUCATION

Bachelor of Engineering, Systems and Computing Engineering (Co-op)

2024–Present

University of Guelph, Guelph, Ontario

- Dean's Honours List – Awarded for achieving an average of 80% in the Winter 2025 semester
- Recipient of Entrance Scholarship of \$2000 for admission average of 93%
- Relevant Courses: Engineering Design 2, Fluid Mechanics, Engineering Systems Analysis

TECHNICAL SKILLS

Embedded & Hardware: Raspberry Pi, STM32, Arduino, C/C++, sensor integration, servo control, VHDL, MATLAB

Mechanical & Design: AutoCAD, SolidWorks, mechanical system integration, system control logic

Software: Python, Java, Kotlin, SQL, Git, Docker, PyTorch

Tools: VS Code, CLion, IntelliJ, PyCharm, Google Cloud Platform, Microsoft Office

EXPERIENCE

Robotics Research Assistant

May 2025 – Present

University of Guelph, Dept. of Electrical and Computer Engineering

Guelph, Ontario

- Designing and building a **predictive assistive robot** for individuals with cognitive disabilities, integrating a Raspberry Pi 5, AI Hat+, STM32G474RE microcontroller, and inertial motion sensor into a unified embedded system.
- Conducted a hardware architecture evaluation comparing the Raspberry Pi 5 and Jetson Orin Nano Super, including add-on modules (AI Hat+2) and microcontroller candidates, selecting optimal components for real-time on-device AI inference.
- Reviewed 11 research papers on assistive robotics and wearable systems, identifying key findings around multi-modal control, computation offloading, and edge AI constraints to guide system design decisions.
- Configured and benchmarked local AI models (Llama 3.2 3B, YOLO) on embedded hardware, evaluating performance trade-offs between cloud and on-device deployment for real-time assistive applications.
- Authored hardware and software requirements documentation and co-wrote a pre-thesis paper covering architecture selection, add-on evaluation, and microcontroller comparison.

PROJECTS

Autonomous Slingshot Rover | *C++*, *AutoCAD*, *Arduino*

- Designed and built an Arduino-based autonomous rover integrating mechanical, electronic, and software subsystems for real-world deployment and targeting.
- Integrated ultrasonic and proximity sensors with servo actuation to enable obstacle detection, autonomous navigation, and automated slingshot firing.
- Developed control logic and system architecture from schematic to physical build, ensuring reliable hardware-software integration under operational conditions.
- Coordinated tasks across a multidisciplinary team, managing design tradeoffs, troubleshooting hardware faults, and meeting project milestones on schedule.

Hydraulic Robot CAD Modelling & Mechanical Analysis | *SolidWorks*

- Reverse engineered a multi-configuration hydraulic robotic machine by creating precise 3D SolidWorks CAD models from physical measurements, capturing real-world geometry across 18+ modelled features.
- Produced detailed 2D engineering drawings (ISO A4) with correct tolerancing and assembled all components into a full 3D CAD animation demonstrating subsystem motion including gears, pistons, and linkages.

Game Boy Emulator | *C*, *SDL2*

- Engineered a fully working Game Boy emulator replicating CPU, memory, PPU, APU, and input subsystems with cycle-accurate instruction decoding for full ROM compatibility.
- Architected a modular system with SDL2 for real-time graphics rendering, enhancing maintainability and enabling scalability for additional emulator features.